2 Multi-Area OSPF with Stub Areas and Authentication

Router R1 Configuration

```
enable
configure terminal

interface loopback1
description Engineering Department
ip address 10.1.1.1 255.255.255.0
exit

interface serial0/0/0
ip address 10.1.12.1 255.255.255.0
clock rate 64000
no shutdown
exit

router ospf 1
network 10.1.12.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.255 area 0
exit

interface loopback1
ip ospf network point-to-point
exit

interface serial0/0/0
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 cisco
exit

end
```

Router R2 Configuration

```
enable
configure terminal

interface loopback2
description Marketing Department
ip address 10.1.2.1 255.255.255.0
exit

interface serial0/0/0
ip address 10.1.12.2 255.255.255.0
no shutdown
exit
```

```
interface serial0/0/1
ip address 10.1.23.2 255.255.255.0
clock rate 64000
no shutdown
exit

router ospf 1
network 10.1.12.0 0.0.0.255 area 0
network 10.1.2.0 0.0.0.255 area 0
network 10.1.23.0 0.0.0.255 area 23
area 23 stub
area 23 stub no-summary
exit

interface loopback2
ip ospf network point-to-point
exit

interface serial0/0/1
ip ospf authentication
ip ospf authentication-key cisco
exit

interface serial0/0/0
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 cisco
exit

end
```

Router R3 Configuration

```
enable
configure terminal

interface loopback3
description Accounting Department
ip address 10.1.3.1 255.255.255.0
exit

interface loopback20
description Connection to another AS
ip address 172.20.200.1 255.255.255.0
exit

interface serial0/0/1
ip address 10.1.23.3 255.255.255.0
no shutdown
exit

router ospf 1
network 10.1.23.0 0.0.0.255 area 23
network 10.1.3.0 0.0.0.255 area 23
area 23 stub
exit

interface loopback3
ip ospf network point-to-point
```

```
exit

interface serial0/0/1
ip ospf authentication
ip ospf authentication-key cisco
exit

end
```

Verification Commands (SAFE to copy)

```
show ip ospf neighbor
show ip route
show ip ospf interface
show ip protocols
```